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| **Fiscal Year:** | FY 2004 | **Task Last Updated:** | FY 03/30/2006 |
| **PI Name:** | Walsworth, Ronald L. Ph.D. | | |
| **Project Title:** | Ground-Based Investigations with the Cryogenic Hydrogen Maser | | |
| **Division Name:** | Physical Sciences | | |
| **Program/Discipline:** | FUNDAMENTAL PHYSICS | | |
| **Program/Discipline--Element/Subdiscipline:** | FUNDAMENTAL PHYSICS--Laser cooled atomic physics | | |
| **Joint Agency Name:** | | **TechPort:** | No |
| **Human Research Program Elements:** | None | | |
| **Human Research Program Risks:** | None | | |
| **Space Biology Element:** | None | | |
| **Space Biology Cross-Element Discipline:** | None | | |
| **Space Biology Special Category:** | None | | |
| **PI Email:** | rwalsworth@cfa.harvard.edu | **Fax:** | FY 617-496-7690 |
| **PI Organization Type:** | UNIVERSITY | **Phone:** | 617-495-7274 |
| **Organization Name:** | Harvard University | | |
| **PI Address 1:** | Smithsonian Astrophysical Observatory | | |
| **PI Address 2:** | 60 Garden Street, Mail Stop 59 | | |
| **PI Web Page:** | http://cfa-www.harvard.edu/ | | |
| **City:** | Cambridge | **State:** | MA |
| **Zip Code:** | 02138 | **Congressional District:** | 8 |
| **Comments:** | | | |
| **Project Type:** | Ground | **Solicitation / Funding Source:** | 96-HEDS-03 |
| **Start Date:** | 02/01/1998 | **End Date:** | 01/31/2005 |
| **No. of Post Docs:** | 1 | **No. of PhD Degrees:** | |
| **No. of PhD Candidates:** | 1 | **No. of Master' Degrees:** | |
| **No. of Master's Candidates:** | 1 | **No. of Bachelor's Degrees:** | |
| **No. of Bachelor's Candidates:** | 1 | **Monitoring Center:** | NASA JPL |
| **Contact Monitor:** | Maleki, Lute | **Contact Phone:** | 818-354-3688 |
| **Contact Email:** | lute.maleki@jpl.nasa.gov | | |
| **Flight Program:** | | | |
| **Flight Assignment:** | | | |
| **Key Personnel Changes/Previous PI:** | | | |
| **COI Name (Institution):** | | | |
| **Grant/Contract No.:** | None | | |
| **Performance Goal No.:** | 1.24 | | |
| **Performance Goal Text:** | To explore the frontiers of physics and unlock the mysteries of the universe obscured by Earth's gravity | | |
| **Task Description:** | The cryogenic hydrogen maser (CHM) operates at 0.5 kelvin, and employs superfluid helium-coated walls to store the masing hydrogen atoms. The CHM may provide frequency stability that is one to three orders of magnitude better than a room temperature hydrogen maser because of greatly reduced thermal noise and larger signal power. In addition, the CHM may be used to study important effects in low temperature atomic physics, e.g., hyperfine-induced hydrogen-hydrogen spin-exchange collisions. To develop the cryogenic hydrogen maser as an improved local oscillator, and to apply hydrogen masers to investigations of fundamental physics such as low temperature hydrogen collisions. Exceptional frequency stability will be required for spacecraft tracking in future deep-space missions, for space-based tests of relativity and fundamental symmetries, and for local oscillators used with new absolute frequency standards such as laser-cooled atomic fountains. | | |

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| **Rationale for HRP Directed Research:** | |
| **Research Impact/Earth Benefits:** | |
| **Task Progress:** | No progress report this period. |
| **Bibliography Type:** | Description: (Last Updated: 07/25/2019) |